

SATO–TATE FAMILIES IN LOCALIZED BOST–CONNES SYSTEMS: OPEN PHASE, NO CONGRUENCE OBSTRUCTION, AND THE SYMMETRIC POWER EXPANSION

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ABSTRACT. Let $E/\mathbb{Q}$ be a non-CM elliptic curve, $a_p = 2\sqrt{p}\cos\theta_p$, and $S_{[a,b]} = \{p : \theta_p \in [a, b]\}$ for $0 \leq a < b \leq \pi$. We settle the three questions posed for the localized system $\mathcal{A}_{\mathbb{Q}, S_{[a,b]}}$.

The phase is open, and necessarily so. By Sato–Tate $S_{[a,b]}$ has positive natural density $\delta = \mu_{\text{ST}}([a, b]) = \frac{1}{\pi}[(b-a) - \frac{1}{2}(\sin 2b - \sin 2a)]$ among the primes, so $\pi_S(x) \sim \delta x / \log x$: the exponent is $\theta = 1$ and the logarithmic power is $k = 1$. By Proposition 1.1 of Paper XXIV the critical series is then $\asymp \delta \log \log x$, divergent, and the symmetry-broken phase is $(1, \infty)$, open. The precise asymptotic is

$$\zeta_{S_{[a,b]}}(\beta) = (\delta + o(1)) \log \frac{1}{\beta-1} \quad (\beta \rightarrow 1^+),$$

with $o(1)$ replaced by $O(1/\log \frac{1}{\beta-1})$ under any effective Sato–Tate error term $O(x/\log^{1+\varepsilon} x)$. This is not an accident of the interval: *a set of positive density can never have a closed phase*, closedness requiring $k \geq 2$, hence density zero with an extra logarithm.

There is no congruence obstruction. The residue $p \bmod m$ is $\det \bar{\rho}_{E,m}(\text{Frob}_p)$, the cyclotomic character, while θ_p is governed by the trace; Serre’s open image theorem makes the mod- m image open, and joint equidistribution of $(\theta_p, p \bmod m)$ follows from the non-vanishing of $L(s, \text{Sym}^k E \otimes \chi)$ on $\Re s = 1$, now unconditional for non-CM $E/\mathbb{Q}$ by Newton–Thorne together with Shahidi. Hence every nontrivial character has a detecting set of positive density, and **the KMS $_{\beta}$ state is unique for all $0 < \beta \leq 1$** . This is the opposite verdict to Theorem 2.2(1) of Paper XXIV of the series, where the Landau and Friedlander–Iwaniec primes are trapped in the kernel class 1 mod 4 and hence never unique — and it is reached by the opposite mechanism to that theorem’s part (2), where the twin and Sophie Germain primes escape obstruction while confined to a single *generating* class. Verified numerically on $y^2 = x^3 + x + 1$ over 6055 primes below $6 \cdot 10^4$: in each of three intervals and each modulus $m \in \{3, 4, 5, 8\}$ every coprime class carries the uniform proportion.

The symmetric powers control the constant, not the transition. Expanding the indicator in the characters $U_k(\cos \theta) = \text{tr Sym}^k$, which are orthonormal for μ_{ST} ,

$$\zeta_{S_{[a,b]}}(s) = \delta \log \frac{1}{s-1} + \sum_{k \geq 1} c_k \log L(s, \text{Sym}^k E) + O(1), \quad c_k = \frac{2}{\pi} \int_a^b \sin((k+1)t) \sin t \, dt.$$

Only the $k = 0$ term, the Riemann zeta function, has a pole at $s = 1$; every Sym^k with $k \geq 1$ is automorphic with $L(1, \text{Sym}^k E) \neq 0$, so contributes $O(1)$. The higher symmetric powers therefore fix the constant in the expansion and the shape of Ξ_{β} , and have no influence whatever on β_c .

We flag one technical point that the L^2 expansion conceals: $\mathbf{1}_{[a,b]}$ is not smooth, so the interchange of $\sum_k$ and $\sum_p$ requires a Beurling–Selberg smoothing, standard in equidistribution but not vacuous.

1. THE THERMODYNAMIC BASELINE

The framework is that of [6], cited by DOI; Papers VI and XXIV of the series, referred to by numeral, are unpublished companions, and every fact used from them is restated or verified in place.

Sources, code and data for the entire series: <https://github.com/Ruqing1963/localized-bost-connes>; this paper is in [papers/XXV-sato-tate/](#).

Proposition 1.1 (Density). *For $0 \leq a < b \leq \pi$,*

$$\delta := \mu_{\text{ST}}([a, b]) = \frac{2}{\pi} \int_a^b \sin^2 t \, dt = \frac{1}{\pi} \left[(b - a) - \frac{1}{2} (\sin 2b - \sin 2a) \right] > 0.$$

By the Sato–Tate theorem for non-CM elliptic curves over $\mathbb{Q}$ [1], $S_{[a,b]}$ has natural density δ in the primes, so $\pi_S(x) \sim \delta x / \log x$.

Theorem 1.2 (Open phase). *$\beta_c(S_{[a,b]}) = 1$, and*

$$(1.1) \quad \zeta_{S_{[a,b]}}(\beta) = \sum_{p \in S_{[a,b]}} p^{-\beta} = (\delta + o(1)) \log \frac{1}{\beta - 1} \quad (\beta \rightarrow 1^+),$$

so that $\zeta_S(1) = \infty$: the symmetry-broken phase is the open half-line $(1, \infty)$. If moreover $\pi_S(x) = \delta \pi(x) + O(x / \log^{1+\varepsilon} x)$ for some $\varepsilon > 0$ — an effective Sato–Tate error term — then the $o(1) \log \frac{1}{\beta-1}$ sharpens to $O(1)$.

Proof. In the notation of Proposition 1.1 of Paper XXIV, $\theta = 1$ and $k = 1$, so $\beta_c = \theta = 1$ and the critical series is $\asymp \delta \int_2^x dt / (t \log t) = \delta \log \log x \rightarrow \infty$. For (1.1), write $E(x) = \pi_S(x) - \delta \pi(x) = o(x / \log x)$ and integrate by parts: $\sum_{p \in S} p^{-\beta} = \delta \sum_p p^{-\beta} + \int_2^\infty x^{-\beta} dE(x)$, the error being $\beta \int x^{-\beta-1} E(x) dx = o(\log \frac{1}{\beta-1})$ from $E = o(x / \log x)$, and $O(1)$ when $E \ll x / \log^{1+\varepsilon} x$, since $\int x^{-2} \cdot x / \log^{1+\varepsilon} x dx < \infty$; finally $\delta \sum_p p^{-\beta} = \delta \log \zeta(\beta) + O(1) = \delta \log \frac{1}{\beta-1} + O(1)$. Closedness would require convergence at β_c , hence $k > 1$ by Proposition 1.1 of Paper XXIV. $\square$

Corollary 1.3 (No positive-density set has a closed phase). *If $S \subseteq P$ has positive natural density then $k = 1$ and the low-temperature phase is open. Closed phases require density zero, and specifically a counting function $\asymp x / \log^k x$ with $k \geq 2$ — which by Remark 1.3 of Paper XXIV means two simultaneous prime conditions. Sato–Tate families, being of positive density, are on the opposite side of that divide from the Sophie Germain family of [6, Thm. 5.5].*

Example 1.4. $\delta = 0.6090$ for $[\pi/3, 2\pi/3]$, 0.0288 for $[0, \pi/6]$, 0.0775 for $[2.4, \pi]$, 0.5 for $[0, \pi/2]$. Empirical densities for $y^2 = x^3 + x + 1$ over the 6055 good primes below $6 \cdot 10^4$: $0.6206, 0.0276, 0.0766$ respectively.

2. NO CONGRUENCE OBSTRUCTION

Lemma 2.1 (The residue is the determinant). *For $m \geq 1$ and $p \nmid mN_E$, $\det \bar{\rho}_{E,m}(\text{Frob}_p) = p \bmod m$, the mod- m cyclotomic character. The angle θ_p is determined by $\text{tr} \bar{\rho}_{E,\ell}(\text{Frob}_p) = a_p$. Thus the residue class and the Sato–Tate angle are carried by the determinant and the trace of the same representation.*

Theorem 2.2 (Joint equidistribution). *Let $E/\mathbb{Q}$ be non-CM. For every $m \geq 1$ the pairs $(\theta_p, p \bmod m)$ are equidistributed in $[0, \pi] \times (\mathbb{Z}/m)^\times$ for $\mu_{\text{ST}} \times \text{unif}$. Consequently, for every interval $[a, b]$ and every residue class r coprime to m ,*

$$\#\{p \leq x : \theta_p \in [a, b], p \equiv r \pmod{m}\} \sim \frac{\delta}{\varphi(m)} \cdot \frac{x}{\log x}.$$

Proof. By Lemma 2.1 and Serre’s open image theorem [3] the mod- m image is open, so the determinant and trace directions are independent in the limit. Analytically, expanding the indicator of $[a, b] \times \{r\}$ in the characters $U_k \otimes \chi$ of $SU(2) \times (\mathbb{Z}/m)^\times$ reduces the statement to the non-vanishing and holomorphy of $L(s, \text{Sym}^k E \otimes \chi)$ on $\Re s = 1$ for all $k \geq 0$ and all Dirichlet $\chi \bmod m$, with a pole only for $k = 0$, χ trivial. Automorphy of $\text{Sym}^k E$ for all k is Newton–Thorne [2]; the non-vanishing on $\Re s = 1$ of the automorphic L -functions and their twists is Jacquet–Shalika [4] and Shahidi [5]. Apply Wiener–Ikehara. $\square$

Theorem 2.3 (High-temperature uniqueness). *For every non-CM $E/\mathbb{Q}$ and every $[a, b]$ with $\delta > 0$, the system $\mathcal{A}_{\mathbb{Q}, S_{[a, b]}}$ has a unique KMS_β state for $0 < \beta \leq 1$, and its KMS_β simplex is $\text{Prob}(G_S)$ for $\beta > 1$. In particular $S_{[a, b]}$ carries no congruence obstruction in the sense of Theorem 2.2(1) of Paper XXIV.*

Proof. By [6, Thm. 3.9] uniqueness at β requires $\sum_{p \in D_\chi} p^{-\beta} = \infty$ for every nontrivial $\chi \in \widehat{G_S}$. For $K = \mathbb{Q}$ such a χ is supported at finitely many places and its detecting set is the union of the classes of $S_{[a, b]}$ outside $\ker \chi$ modulo some m ; by Theorem 2.2 that set has density $\delta(1 - 1/[\cdot]) > 0$ in the primes, so its Dirichlet series diverges for $\beta \leq 1$ by Theorem 1.2. For $\beta > 1$ the whole of $\zeta_S(\beta)$ converges, so $\Xi_\beta = \widehat{G_S}$. $\square$

Remark 2.4 (Numerical verification). For $y^2 = x^3 + x + 1$ (non-CM), the 6055 good primes below $6 \cdot 10^4$:

$[a, b]$	$m = 3$	$m = 4$	$m = 5$	$m = 8$
$[\pi/3, 2\pi/3]$	.500, .500	.502, .498	.246, .250, .254, .249	.253, .248, .248, .250
$[0, \pi/6]$	.431, .569	.449, .551	.275, .222, .269, .234	.228, .251, .222, .299
$[2.4, \pi]$	.478, .522	.496, .504	.259, .274, .235, .233	.246, .265, .250, .239

against uniform values $1/2$ and $1/4$. Every coprime class is occupied at roughly the uniform proportion; the fluctuations in the two sparse intervals are consistent with the sample sizes (167 and 464 primes). Contrast Remark 2.3 of Paper XXIV, where the twin and Sophie Germain families occupy the single class $2 \bmod 3$ — a generating class, so no obstruction — and the Landau and Friedlander–Iwaniec families the single kernel class $1 \bmod 4$, which does obstruct.

Remark 2.5 (Why this family passes the test). The obstructed families of Paper XXIV — Landau and Friedlander–Iwaniec — are defined by a condition that *involves* the residue, p being a value of a polynomial, so the condition and the residue are not independent, and the dependence lands in the kernel. (The twin and Sophie Germain conditions are also residue-entangled, but land in a generating class and escape.) Here the defining condition involves a_p , and by Lemma 2.1 the residue is the determinant while a_p is the trace of the same Galois representation. Open image is exactly the statement that these two coordinates are independent. The distinction is structural, not a matter of the density being larger.

3. THE SYMMETRIC POWER EXPANSION

Proposition 3.1 (Chebyshev coefficients). *The functions $U_k(\cos t) = \sin((k+1)t)/\sin t$, $k \geq 0$, are orthonormal in $L^2(\mu_{\text{ST}})$ and satisfy $U_k(\cos \theta_p) = \text{tr Sym}^k(U_p)$ where $U_p = p^{-1/2}\rho_E(\text{Frob}_p)$. Writing $\mathbf{1}_{[a, b]} = \sum_{k \geq 0} c_k U_k$ in $L^2(\mu_{\text{ST}})$,*

$$c_0 = \delta, \quad c_k = \frac{2}{\pi} \int_a^b \sin((k+1)t) \sin t \, dt = \frac{1}{\pi} \left[\frac{\sin kt}{k} - \frac{\sin(k+2)t}{k+2} \right]_a^b \quad (k \geq 1).$$

Theorem 3.2 (The expansion). *For $\Re s > 1$,*

$$(3.1) \quad \zeta_{S_{[a, b]}}(s) = \delta \log \frac{1}{s-1} + \sum_{k \geq 1} c_k \log L(s, \text{Sym}^k E) + O(1),$$

the series over k summed by truncation as in Remark 3.3, and every term with $k \geq 1$ is $O(1)$ as $s \rightarrow 1^+$, since $\text{Sym}^k E$ is automorphic (Newton–Thorne [2]) and $L(1, \text{Sym}^k E) \neq 0$ (Shahidi [5]). Hence the pole, the critical exponent $\beta_c = 1$ and the divergence at $\beta = 1$ all come from the $k = 0$ term, the Riemann zeta function; the symmetric powers of higher order determine the constant and the fine structure of Ξ_β but not the transition.

Proof. $\log L(s, \text{Sym}^k E) = \sum_p \text{tr} \text{Sym}^k(U_p) p^{-s} + O(1)$ for $\Re s > 1/2$, the prime-power terms converging by Deligne's bound. Substitute the expansion of the indicator and use $\log \zeta(s) = \log \frac{1}{s-1} + O(1)$ for the $k = 0$ term. $\square$

Remark 3.3 (The smoothing is not vacuous). $\mathbf{1}_{[a,b]}$ is not continuous, so the Chebyshev series converges in $L^2(\mu_{\text{ST}})$ but not uniformly, and the interchange of $\sum_k$ with $\sum_p$ in (3.1) is not justified term by term. The standard remedy is to bracket the indicator between Beurling–Selberg majorants and minorants of bounded degree, which introduces an error $O(1/K)$ from truncating at $k \leq K$ and is then optimized against the error term in the Sato–Tate count. We note that the truncation must also absorb the k -dependence of the individual $O(1)$'s: both the prime-power tails ($|\text{tr} \text{Sym}^k| \leq k + 1$) and the size of $\log L(s, \text{Sym}^k E)$ near $s = 1$ grow with k through the analytic conductor, so the constants are not uniform and the series over k converges only through the bracketing, not term by term. The formula is correct; the proof of the interchange is the usual one and is not automatic.

Example 3.4 (Coefficients). For $[a, b] = [\pi/3, 2\pi/3]$: $c_0 = 0.608998$, $c_1 = 0$, $c_2 = -0.413497$, $c_3 = 0$, $c_4 = 0.137832$, $c_5 = 0$, $c_6 = 0.068916$. The odd coefficients vanish by the symmetry of the interval about $\pi/2$; $\text{Sym}^2 E$ carries the leading correction, which is the familiar statement that the second moment of a_p is the first place a symmetric interval is sensitive to the curve.

4. ASSESSMENT

Question	Answer	Reference
$\beta_c(S_{[a,b]})$	1	Thm. 1.2
phase at $\beta = 1$	OPEN $(1, \infty)$	Thm. 1.2
$\zeta_S(\beta)$, $\beta \rightarrow 1^+$	$\delta \log \frac{1}{\beta-1} + O(1)$	Thm. 1.2
positive density $\Rightarrow$ closed?	never	Cor. 1.3
joint equidistribution	holds, unconditionally	Thm. 2.2
congruence obstruction	none	Thm. 2.3, Rem. 2.4
unique KMS_β for $\beta \leq 1$	yes	Thm. 2.3
Sym^k controls β_c	no; only the constant	Thm. 3.2
term-by-term interchange	needs smoothing	Rem. 3.3

No no-go theorem is warranted here, and that is itself the finding. The Sato–Tate intervals pass the congruence test of Paper XXIV by *equidistribution* — the clean mechanism — where the twin and Sophie Germain families pass it, by that paper's revised Theorem 2.2(2), through the opposite extreme of confinement to a generating class. The reason here is structural: the defining condition is a constraint on the *trace* of the Galois representation, while the residue class is its *determinant*, and Serre's open image theorem [3] is precisely the independence of the two. The obstructed families of Paper XXIV — Landau and Friedlander–Iwaniec — fail because their defining conditions are congruence conditions in disguise, landing in the kernel.

The price is that the phase is open. Positive density forces $k = 1$ and hence divergence at the critical point, so these systems cannot exhibit the closed low-temperature phase that [6, Thm. 5.5] found for the Sophie Germain family. By Corollary 1.3 the two desiderata — a closed phase and an unobstructed high-temperature phase — are in tension: the first needs density zero with two prime conditions, the second is easiest for sets defined by trace conditions, which have positive density. By the revised Theorem 2.2(2) of Paper XXIV the twin, Sophie Germain and Chen families all combine a closed phase with the absence of obstruction, Chen unconditionally on the divergence side; those remain the only such families we know.

Two questions. Can one intersect a Sato–Tate interval with a sparse condition to obtain $k = 2$ while keeping joint equidistribution, for instance $\{p : \theta_p \in [a, b], p + 2 \in P_2\}$ — the Chen

condition contributing the extra logarithm and the trace condition remaining independent of it? And second, the anomalous sets of Theorem 5.3 of Paper VI were built by letting $[a, b]$ shrink with p ; by Theorem 1.2 any *fixed* interval gives $\beta_c = 1$ with an open phase, so the whole of the phase-structure interest in Sato–Tate families lies in shrinking intervals, which is where Paper VI placed it.

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